The Noita Eye Messages

Report for the community: what we found, what it rules out, and a call for cribs — 30 September 2026

Summary

This is an independent analysis of the nine Eye Messages. We rebuilt the ciphertext from the game's own pixels, checked it byte for byte against the copy embedded in the game executable, and then tested one family of cipher mechanisms after another against it. Nothing was used until it had been reproduced from primary data. Every instrument also had to pass synthetic "known-answer" gates before its verdict on the real messages was read.

The cipher is **not solved**. The main results:

- **Provenance closed.** The pixel transcription equals the eye data inside `noita.exe` (January 2025 build), character for character. The game ships the ciphertext and nothing else.
- **Period-4 is a duplication artifact.** The distance-4 excess disappears once each repeated event is counted once.
- **Large-state models don't fit the resyncs.** A group autokey over S83 or A83 (a per-letter deck shuffle) should essentially never realign; the messages do it four times.
- **Leading model.** A group autokey over C83:C41 (affine maps) with small per-letter steps. It reproduces most of the fingerprint, but it is a model, not a solution.
- **Key recovery needs a crib.** The ciphertext alone does not pin the key, so we built a crib tester. We are asking for crib candidates (section 6).

How you can help: send guesses for the two shared openings — 24 letters shared by east-1, east-2 and west-1, and 20 letters shared by east-4, east-5 and west-4. Finnish is welcome and may be more likely. Section 6 has the format.

1. Data and provenance

Every eye was located by exact template match. Of the 36 possible digit-significance orders, exactly one gives a contiguous 0–82 alphabet of 83 letters; it is the standard community reading. We then extracted the eye data straight from the current `noita.exe` (64-bit constants, base-7 packed). The 3,194-character stream is identical to our pixel transcription, and there are no other encoded runs in the binary: no hidden tenth message, no west-5.

A full sweep of `data.wak` — a keyword search in English and Finnish over all 14,745 files, plus a pixel scan of 9,046 images for grids of eyes — found no decoder, no plaintext table and no extra mural.

Fact	Value
Corpus	9 messages, 3,108 eyes, 1,036 letters; lengths 99/118/137/119/114 (east), 103/102/124/120 (west)
Alphabet	83 letters, 0–82 contiguous; reading order unique among 36
Adjacent doubles	**0** observed where ~12 are expected: built into the mechanism
Openings	Position 0 differs in all 9. Positions 1–2 are the same in all 9 (66, 5). {east-1, east-2, west-1} share 1–24; {east-4, east-5, west-4} share 1–20
Resyncs	4: two messages differ for 1–4 letters, then become identical again (e.g. east-1/west-1 realign for 13 letters at position 37)

2. What is ruled out

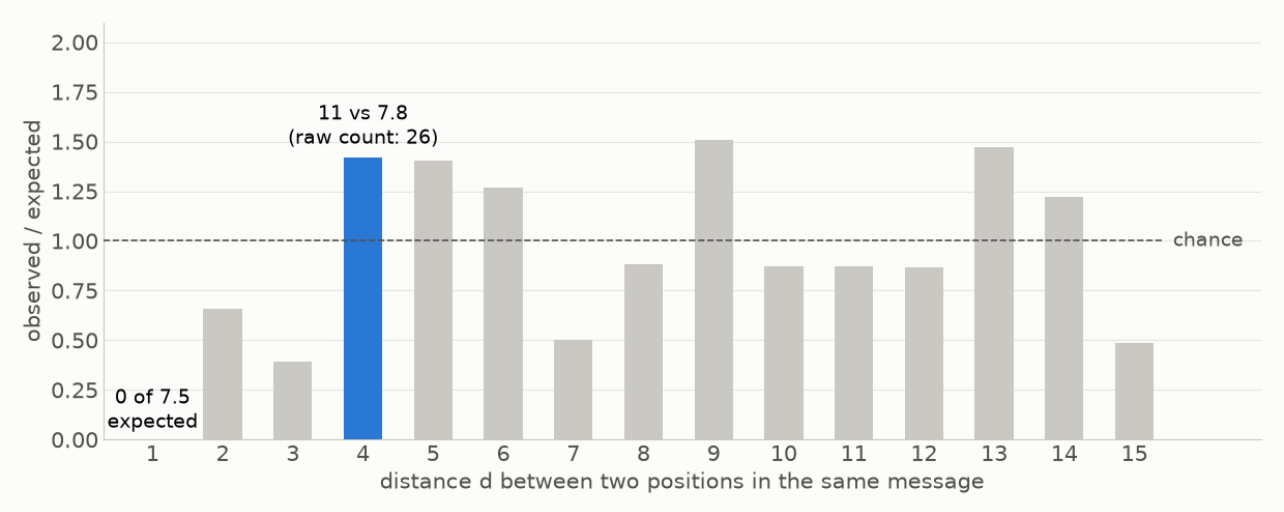
Each verdict below comes from an instrument that first had to crack, or correctly reject, synthetic corpora of known construction.

Hypothesis	Verdict	Deciding evidence
Additive / cyclic-group walk (“cut deck”, incrementing wheel)	Refuted	Isomorph equations force letter merges, for every prime modulus 59–127. Agrees with the community’s C83 chaining result
Collision-skip walk (“step again if it would double”)	Refuted	Rare skips would still crack; dense skips cannot make the observed isomorphs
Re-keying locked to rows	Refuted	The L=33 isomorph east-5[33:66] ~ west-4[34:67] runs across a row-pair boundary in both copies
Period-4 additive key index; pure period-4 keystream	Refuted	Isomorph offsets spread over all residues mod 4 with non-identity σ
Single progressive rotor (σ = Tδ)	Not supported	0 of 4 independent edges fit, against 42 of 42 on the control
Large-state autokey (S83 / A83 GAK)	Refuted	Cannot re-collide into a 13-letter identical run after a differing patch; the corpus does so 4 times
K-state machines (K = 8–30), hybrid registers	Refuted	Their isomorphs are plain repeats; walk dimension and resync are wrong
Pupil geometry in σ	Null	Digit preservation (msd/mid/lsd) is at shuffled-null levels
Decoder or key shipped with the game	Refuted	Full sweep of data files and the executable

3. Period-4 is an artifact, and there are new isomorphs

The distance-4 excess (26 equal-letter pairs four apart, against ~12 expected) was the main periodicity signal, for the community and for us. Most of those 26 hits are **copies of the same event**: family members share their openings, and isomorph partners repeat each other. We merged identical same-position letters and every aligned isomorph position with union-find, then counted each recurrence once.

Figure 1. Repeated symbols by distance, each event counted once



Deduplicated counts against a Poisson expectation. d4 = 11 against 7.8 (p = 0.16) is level with d5, d9 and d13: there is no period. What survives is near-repeat depletion: d1 = 0, with d2 and d3 low. Source: `data/dedup_recurrence.json`.

We also found and retracted an error in our own earlier notes: a second “period-4” observable we had cited was a transcription mistake. There was only ever one, and it is the artifact above.

An exhaustive diagonal scan (every pair of messages at every offset) found 28 real isomorph windows; shuffled and reversed controls give none. **16 of the 28 are new**, among them:

- east-4[68:102] ~ west-4[71:105], length 34
- east-5[66:99] ~ west-4[68:101], length 33
- **east-3 is linked for the first time**: it ties to the family-B tail and to west-1 (east-3[86:109] ~ west-1[55:78]).

Counting alphabets: the nine occurrences in the isomorph webs need **at least 8 distinct substitution alphabets**, with exactly one proven reuse, inside the east-1/west-1 resync region.

4. Leading model: gak41 with short steps

Of the six transitive groups in the community's group-autokey classification, C83:C41 was the only one nobody had ruled out. It has 41 multipliers, so its state is small:

- state (b, a), affine update $x \to b \cdot x + a \bmod 83$, with b in the order-41 subgroup;
- output c = D[a]; $a \neq 0$ rules out doubles structurally;
- **per-letter additive steps are small (1..27)**, and **2–3 "twist" letters** change b.

We scored candidates over five tournament rounds against the real fingerprint: long exact isomorphs, walk dimension 1, non-commuting σ, zero doubles, d2/d3 depletion, resync rate and IoC. This model averages **about 5.5 of 7**, and is the first candidate to reproduce walk dimension 1 at all. The short steps are what produce the soft d2/d3 depletion.

Candidate (200 seeds)	Isomorphs	Walk dim 1	Noncomm.	Zero doubles	Mean score
Short steps K=27, 3 twists	1.00	0.69	0.84	1.00	5.53 / 7
Short steps K=27, 5 twists	1.00	0.64	0.92	1.00	5.43 / 7
+ plaintext resets (q z j x)	0.99	0.52	0.76	0.96	5.25 / 7
State-driven twists (12 variants)	0.02–0.97	≤ 0.16	≤ 0.62	1.00	2.7–4.6 / 7
Coarse-step autokey	1.00	0.54–0.58	0.81–0.92	0.03–0.06	3.7 / 7

Pass rates per test, from tournament round 5 (`data/tournament5_report.json`).

On resync: with short 1–4-letter plaintext edits, like the real mismatch islands, even this lossless autokey realigns by coincidence often enough. So the four resyncs **do not prove** the state update loses information. Resets and state-driven twists were tried as lossy updates, and both break the isomorphs.

The remaining misses are noise in the walk-dimension test and a resync rate that depends on how the real plaintexts differ, which is unknown.

5. Key recovery: where it stands

Method	Result	What it means
Affine fit on the isomorph web	Inconclusive	Once the two short excursions (east-1/west-1 at 25–28 and 33–36) are cut out, the web fits gak41, but with a null space of dimension 8. It neither refutes the model nor pins the key
Blind annealing for the hidden order π	Gate fails	The true key scores 0 and every reachable optimum ≥ 305. The basin is only ~5 transpositions wide, so the search ends in decoys
Crib tester (new)	Sound, low power	Synthetic: 0 of 72 wrong cribs accepted, 0 of 12 true cribs rejected. Accepted true cribs recover π on 67–73 symbols. Most runs hit the search cap

The crib tester turns a guessed plaintext for a shared opening into exact GF(83) equations and solves them together with the isomorph equations. It is the first method in the project that reaches the key. Longer cribs (~20 letters) work best, and each costs about 72 minutes on a 16-core machine.

> On real data, an **accept** would be decisive: decrypt all nine messages and read them. A **reject** rules the crib out. A **capped** run means nothing either way.

6. Call for cribs

The tester needs guesses for the two shared openings. Position 0 is a per-message header outside the chain, so it is not included.

Opening	Messages	Positions	Letters
A	east-1, east-2, west-1	1–24	24
B	east-4, east-5, west-4	1–20	20

Format: one crib per line, `A<TAB>phrase` or `B<TAB>phrase`. Letters only; each phrase is tried both with and without spaces as symbols, and cut to 24 or 20 letters. **Finnish is welcome and may be more likely** — Nolla Games is a Finnish studio, and ä/ö are handled.

Good sources are in-game text, lore, already-cracked Noita puzzles, and boilerplate that fits the family structure. Opening A repeats 4 symbols internally, so a guess needs matching repeated letters in the same places; those checks reject some guesses instantly.

Caveats. The tester assumes the gak41 short-step model, allows at most one twist letter inside the crib, and has low power. Every result in this report comes with its script and a gate that has to pass before the real-data verdict counts.